

(Digital) Video



Frame rate and resolution

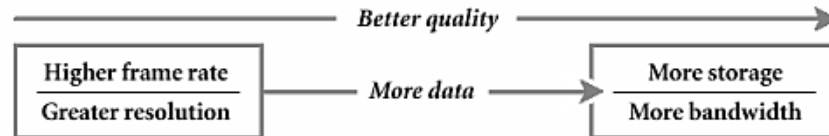
Frame rate

Number of frames per second

- At least 10 f/s are necessary to perceive a fluid movement
- Cinema: 24 fps
- Television: 25/30 fps (Europe / USA and Japan)

Resolution

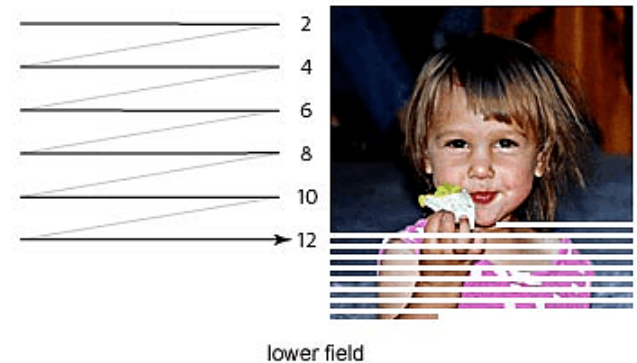
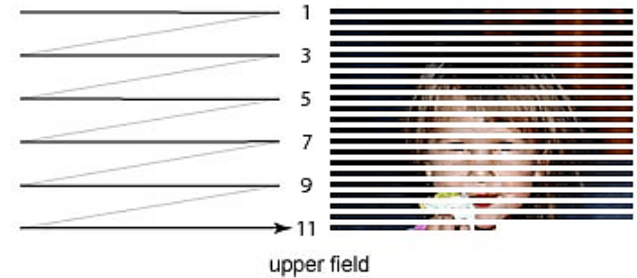
Size (in pixels) of frames (width x height)



Interlaced video

Each frame is composed of two *fields*, displayed in two phases

- (Analog) PAL, SECAM → 25 fps → 50 fields/sec
- (Analog) NTSC → 30 fps → 60 fields/sec



Interlaced vs. non-interlaced video

Non-interlaced video

Typical of computer screens

Also called *progressive scan*

Each frame is displayed “as a whole”

E.g., monitor/graphic card with 75 Hz *refresh frequency* → whole image displayed 75 times per second



Important: *interlaced/non-interlaced* are not referred to the medium or format

E.g., digitized PAL video (int.) can be displayed by a computer monitor (non-int.), and digitized 35mm video (non-int.) can be displayed by an old CRT television set (int.)

Color coding

YCC color...

Color model used for analog video broadcasting

The basic principle is still that of RGB

The broadcast signal is formed of a luminance component (Y) and of two color components (CC)

Y → the only signal necessary for B/W TV

Principle

Y = luminance (depends on R, G, and B)

$$C_b \approx B - Y$$

$$C_r \approx R - Y$$

... YCC color

Practically, Y , C_b , and C_r are obtained as linear combinations of the RGB components

E.g., PAL system (YUV)

$$\begin{cases} Y = 0.299 R + 0.587 G + 0.114 B \\ U = -0.147 R - 0.289 G + 0.436 B \\ V = 0.615 R - 0.515 G - 0.1 B \end{cases}$$

NTSC system (YIQ)

$$\begin{cases} Y = 0.299 R + 0.587 G + 0.114 B \\ I = 0.596 R - 0.274 G - 0.322 B \\ Q = 0.212 R - 0.523 G + 0.311 B \end{cases}$$

Green contributes more than red and blue to luminance...

The method is also used in some graphic formats

E.g., in the old Kodak PhotoCD format

Analog video formats

Types of connections between devices

- **Composite**

The YCC video signal is conveyed through a single cable; acceptable quality

- **S-Video**

The luminance (Y) and the C_b and C_r components are transmitted through two distinct cables (included in a single cable); good quality

- **Component**

The luminance (Y) and the C_b and C_r components are transmitted through three distinct cables; high quality



Tape format	Video format	Quality	Use
VHS, 8mm	Composite	Acceptable	Domestic
S-VHS, Hi-8	S-Video	Good	Advanced domestic, industrial
BetaSP	Component	Very good	Industrial, television

Analog broadcast standards

PAL, NTSC, SECAM



Broadcast formats	Countries	Horizontal lines	Frame rate
PAL	Europe (almost all), China, South America, Australia	625	25 fps
SECAM	France, Middle East, most of Africa	625	25 fps
NTSC	USA, Canada, Japan, Korea, Mexico	525	29.97 fps

The tape format is independent of the standard used

E.g., a VHS tape can be used for PAL, NTSC, and SECAM video

Video acquisition

From analog devices

- E.g., old analog camcorders or video recorders (VHS, VHS-C, S-VHS, 8mm, Hi-8, Beta-SP)
- It is necessary to use a *capture device* to digitize the analog signal
After digital processing, the video can be converted into an analog format again

From digital camcorders/smartphones

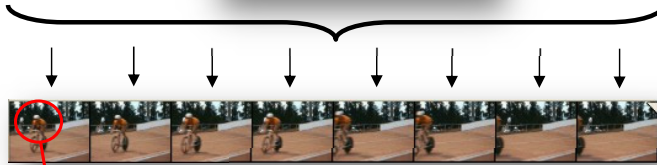
The connection is direct

Typically, through a USB – or, in the past, *FireWire* (IEEE 1394) – connection

Video – A/D conversion process



Analog source (e.g., VHS video recorder)



Time sampling (e.g., 15 frames per second)



Spatial sampling (e.g., 320x240 pixels)

Quantization (e.g., 24 bits to define the color of each pixel)

Video compression...

Video digitization (almost) always implies some compression

Without compression, files would be huge...

E.g., 640x480 video, 25 fps, RGB color (i.e., 24 bits)

1 second → $640 \cdot 480 \cdot 3 \cdot 25 = 23,040,000$ bytes ≈ 22 MB (*data rate*)

1 minute → $\approx 22 \cdot 60 = 1,320$ MB ≈ 1.3 GB

1 hour → $\approx 1.3 \cdot 60 = 78$ GB

Intra-frame compressions

Each frame is compressed separately

Inter-frame compressions

Compression occurs by analyzing differences between frames

... video compression...

Codec

= Compressor/decompressor, implements the compression/decompression algorithm

Refers to both hardware (e.g., digital camcorders or digitizing devices) and software

- There are many types of codecs

E.g., some use a fixed data rate, others compress each frame depending on its content

Format	Resolution	Type of compression	Data rate	Use
MJPEG	720 x 486	Intra-frame	0.5 – 25 MB/s	General
MPEG-1	352 x 240	Intra-frame	0.01 – 0.06 MB/s	CD-Rom, Web
MPEG-2	720 x 480	Intra- and inter-frame	0.01 – 2 MB/s	DVD, digital TV
DV	720 x 480	Intra-frame	3.5 MB/s	Domestic, industrial, broadcast
D1	720 x 486	Uncompressed	25 MB/s	Broadcast

... video compression...

Important: Codec $\neq$ Format

The same video format can usually support several codecs

E.g., some codecs supported by AVI and QuickTime formats:

- AVI: Indeo Video 5.06, Microsoft RLE, Microsoft Video 1, Cinepak, DivX, ...
- QuickTime: Indeo, Motion JPEG A, Animation, DV PAL, H.264, ...

Possible confusion between format and codec

E.g., MPEG2 can be considered both a format and a codec

... video compression...

Spatial compression

- Reduction of frame size (obvious, lower spatial sampling rate)
E.g., with 320x240 frames instead of 640x480 frames, the file size is reduced to 1/4
- Reduction of the number of bits used to code color (obvious, lower quantization)
 - The human eye is much more sensitive to luminance variations than to color changes; groups of pixels having similar color components can be forced to the same color components
 - Normally, image quality decreases as compression increases

... video compression...

Time compressions

- Reduction of frame rate (obvious, lower time sampling)
E.g., with 13 fps instead of 25, the file size is reduced by (about) 1/2
- Coding of changes between frames rather than complete images
 - E.g., in a video containing a still person who is talking, relevant changes will be only in the head area
 - Each frame that is very different from the previous one becomes a *keyframe* and is completely coded

Lossless and lossy compressions

- *Lossless*: all the original information is preserved (high data rates)
- *Lossy*: loss of information (lower data rates)

... video compression

Symmetric and asymmetric compressions

- *Symmetric*: compression time (during video production) is similar to decompression time (by the video player)
- *Asymmetric*: compression and decompression have significantly different durations
 - Fast compression → Simpler algorithms
 - Slow compression → Smaller files
 - Fast decompression → Useful for not-very-powerful computers

In general, low decompression times require high compression times

History: the Digital Video format

DV technology...

DV tapes

Used in DV camcorders (mini-DV format is/was the most common)



DV compression

The most common (DV25) uses/used a 25 Mb/s fixed data rate

DV camcorders

For both consumer and professional use;
employ/employed DV tapes



... DV technology...

Advantages of digital video (compared to analog video)

- Better image and sound
- No quality loss
- No need for digitizing devices
- Better tapes and smaller camcorders

Camcorder-PC connection

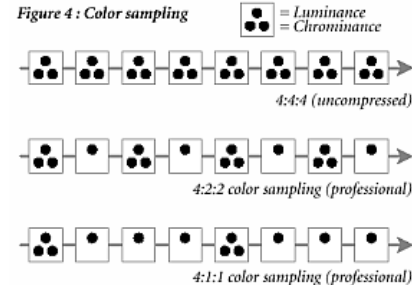
Typically, by means of USB3 or (especially in the past) IEEE 1394 connection, also known as *FireWire* (Apple) and *i.LINK* (Sony)

... DV technology

DV25 compression

- Fixed 5:1 ratio
- Data rate = 25 Mbits/sec = 3.125 MB/sec + control data and audio data $\approx$ 3,6 MB/sec
→ 1 hour $\approx$ 13 GB
- Based on the reduction of the color sampling rate (audio not compressed)

Since the eye is more sensitive to luminance variations than to color changes, luminance is sampled at each frame, while color is sampled every 2 (4:2:2, 50 Mbps) or 4 (4:1:1, 25 Mbps) frames



MPEG Technologies

MPEG = Motion Picture Expert Group

- Standard defined in January 1988
- Initially, it required costly devices

MPEG-1

Created for CD-ROM video – Typical data rates ~ 1.5 Mb/sec Resolution: 352x288 (PAL) and 352x240 (NTSC)

MPEG-2

Suitable for broadcast – Typical data rates ~ 1 MB/sec – Resolution: 704x576 (PAL) and 704x480 (NTSC) - Supports 16:9 formats

MPEG-2

Used for DVDs and digital TV

“High” (in the past) quality, low data rate

MPEG-2 as a universal format for digital video?

Both intra- and inter-frame algorithms → High compression times → In the past, they were not suitable for real-time recording and editing using cheap devices (the DV format was more convenient)

Three kinds of frames

- I: *intra-frame*, similar to DV frames
- P: *predicted frame*, based on previous frames
- B: *bi-directional frame*, based on both previous and next frames

Typical sequence: *I-P-P-P-P-B-B-B-B-P-B-B-B-P-I-P-P-P-P-B-B-B-B-P-I-P-P-P-B-B*

Beyond MPEG-2

MPEG-1 and MPEG-2 encode “ordinary” video and audio content...

MPEG-4

Lower data rates

Higher error robustness, integration of different “objects” in the same container, and interactive content

MPEG-7

Defines a “framework” for the **description** of content

Also in a very detailed way, thus facilitating indexing

MPEG-21

Defines an “interoperable multimedia framework”

Integration of different multimedia elements and fundamental role of the user

MPEG-4

Ref. <https://mpeg.chiariglione.org/sites/default/files/files/standards/docs/w4668.zip>

MPEG-4, formally called *ISO/IEC 14496*, was finalized in October 1998

It became an International Standard in the first months of 1999

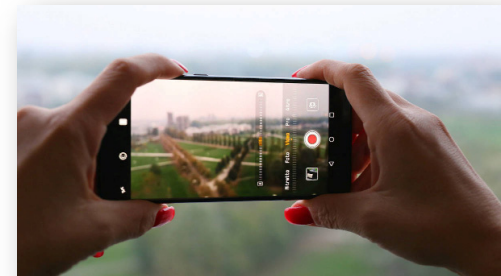
MPEG-4 builds on the proven success of three fields:

- Digital television
- Interactive graphics applications (synthetic content)
- Interactive multimedia (World Wide Web, distribution of, and access to content)

MPEG-4 provides the standardized technological elements enabling the integration of the production, distribution, and content access paradigms of the three fields

H.264/AVC (Advanced Video Coding)...

- MPEG-4 codec
- Also known as “MPEG-4 Part 10”
- Currently, probably the most used codec
- Suitable for High Definition video (Full HD and 4K)
- Codec of *Blu-Ray* discs
- Natural successor of MPEG-2

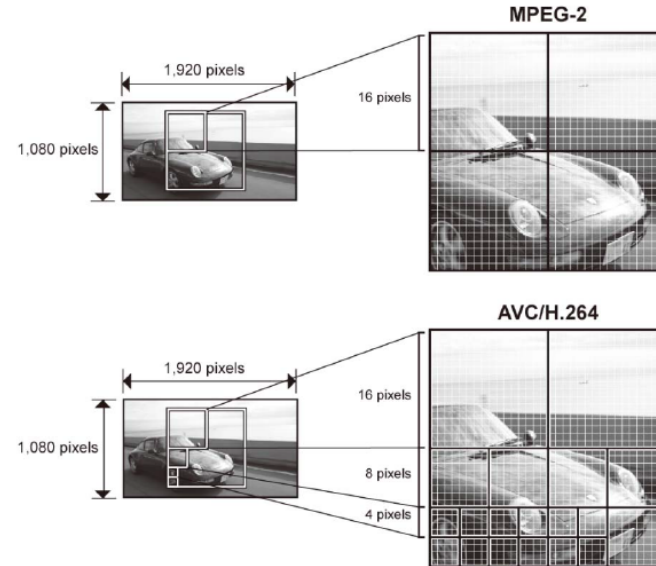


MPEG-4 format

... H.264/AVC...

High-quality image achieved
through higher-precision
“motion- prediction” blocks

A conventional MPEG-2 block
unit used for motion prediction is
16x16 pixels, while in H.264 it
can be 4x4 pixels



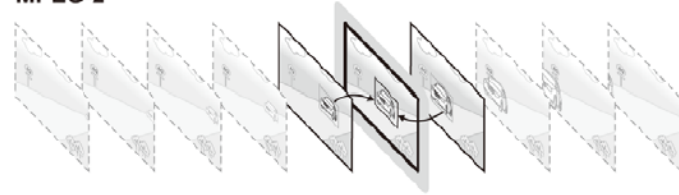
Compared to MPEG-2 compression, H.264 divides the image into smaller pixel blocks for higher picture quality

... H.264/AVC

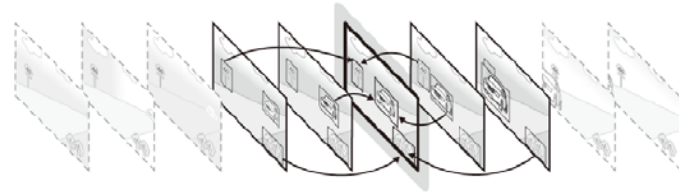
Interframe compression

- MPEG-2: uses one frame before and one frame after
- H.264: uses four frames at a time

MPEG-2



AVC/H.264



Some web video formats

MOV (QuickTime)

- Developed by Apple (but also supported by Windows)
- Supports millions of colors and CD-quality audio
- Very common and versatile

AVI = Audio Video Interleaved

- Windows format
- Supports millions of colors and CD-quality audio
- Very common and very versatile

High-definition video

HD formats...

Resolution	Aspect ratio	Interlaced frames per second (fps)	Progressive fps
1280 x 720	16:9		23.976, 24, 29.97, 30, 59.94, 60
1920 x 1080	16:9	25 (50i), 29.97 (59.94i), 30 (60i)	23.976, 24, 29.97, 30

720p (HD ready)

1280 x 720, 921,600 pixels, progressive

1080i

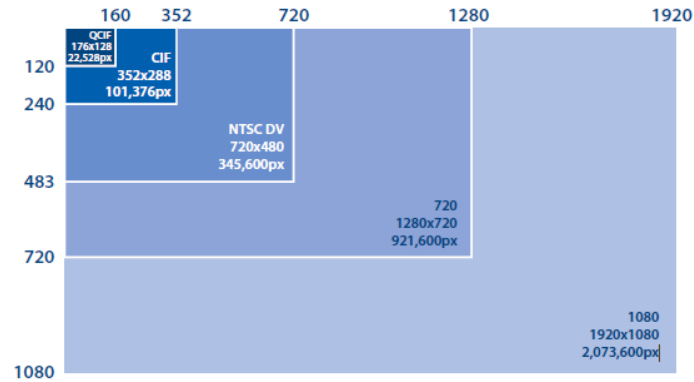
1920 x 1080, 2,073,600 pixels, interlaced

1080p (Full HD)

1920 x 1080, 2,073,600 pixels, progressive

High-definition video

... HD formats



4K video

The frame height is twice that of Full HD



DCI 4K (Digital Cinema Initiative 4K):
width = 4,096

UHDTV (Ultra High Definition
TeleVision 4K): width = 3,840

Streaming

(Video and Audio) Streaming

Allows watching videos and listening to music while files are being downloaded from the Internet

As soon as a portion of the file arrives,
it is played



“Pure streaming” vs. “progressive download”

Pure streaming (*real-time*):

- Uses protocols different from HTTP(S) (e.g., RTSP)
- Supported only by specific formats (e.g., MP3 and Real Audio for audio, and QuickTime and Windows Media Video for Video)
- Needs specific servers
- Downloaded files are not saved locally

Progressive download

- Behavior similar to that of pure streaming
- Uses the HTTP/HTTPS protocol (does not require special servers)
- Usually needs less bandwidth

Phases in professional video creation

1. Pre-production

Outline (what to do), *script* (dialogues, notes, etc.), *storyboard* (“strip cartoons” describing key scenes), *budgeting* (expenditure forecast), *casting* (selection of actors and actresses), *locations* (selection of places), ...

2. Production

Actual shooting

3. Post-production

“Composition” and editing of the acquired videos (editing, titles, special effects, etc.)

Non-linear editing...

Refers to video/audio editing carried out through a computer (using specific software)

As opposed to *linear editing*, where scenes from different sources are mixed on a single support (e.g., tape)

Allows for operating on a single frame

Exact editing

“Easy” creation of special effects

Frames can be modified in many ways (as well as audio tracks)

... non-linear editing

Online editing

The final video is built by directly operating on the original (high-quality) clips

- In the past, powerful and expensive workstations were necessary
- Current computers are often sufficiently powerful to process high-quality video

Offline editing

A preliminary video is built using low-quality clips, and only at the end (after the editing) is it converted to a high-quality video

Especially useful if the computer is not powerful

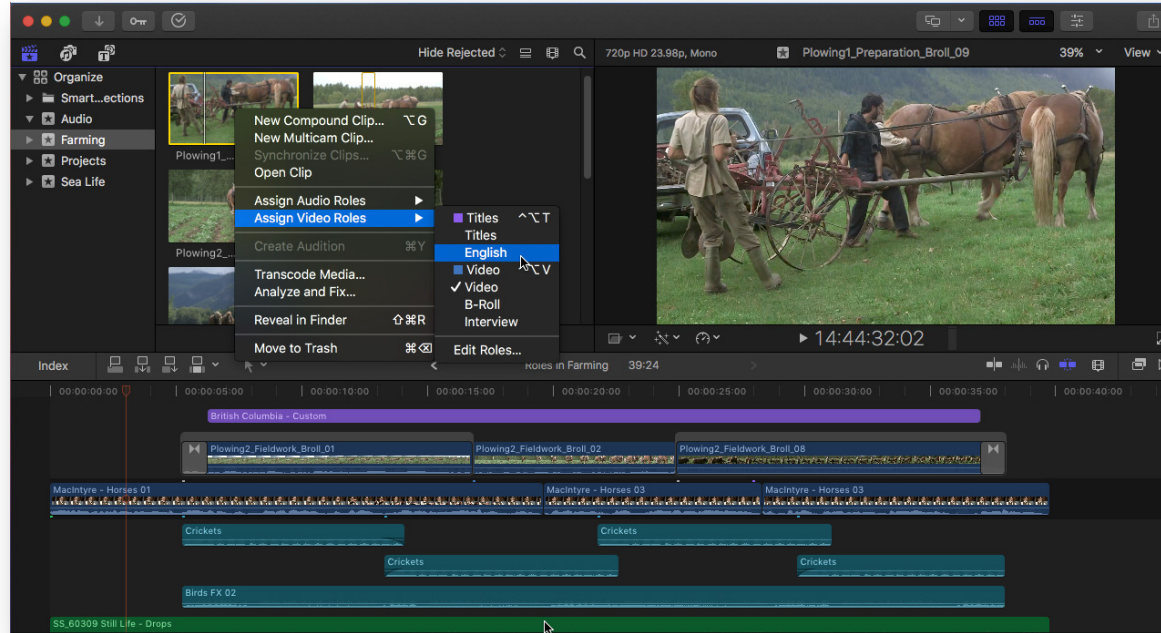
Non-linear editing software2.6

Adobe Premiere



Non-linear editing software – Widespread commercial products

Apple Final Cut



Non-linear editing software – Widespread commercial products

DaVinci Resolve (free version available)

